

Project Proposal: Predictive Analytics for Telecom Customer Retention

Customer acquisition is a massive financial sinkhole. Why pour money into marketing when your existing base is quietly walking out the back door? That's the core problem this project tackles. We aren't just guessing why users leave. We're predicting it.

I selected the IBM Telco Customer Churn dataset from Kaggle to build a binary classification machine learning model. The raw file tracks 7,043 distinct customers across 21 specific features. We have demographics, account configurations, and exact behavioral billing metrics. Most importantly, it gives us a hard, binary outcome variable: did the customer churn, yes or no?

Evaluating retention requires clean, rigorously audited data. Right now, this raw dataset isn't ready for an algorithm. Here is exactly what I will accomplish during this exploratory data analysis (EDA) and preprocessing phase.

- First, strict data isolation. I'll split the data 70/30 programmatically in Python before executing a single transformation. The test set goes into a vault. Zero data leakage allowed.
- Then comes the heavy lifting. I'll strip out useless identifiers like customer IDs. Hidden blanks in the TotalCharges column will be mathematically coerced into nulls and imputed using the training set median. Categorical text, like payment methods and internet service types, can't be read by a model. I'll transform these into numeric matrices using Label Encoding and One-Hot Encoding.

- The continuous variables need leveling. I'll apply MinMax scaling so raw features like tenure and monthly charges don't disproportionately overwhelm the algorithm.
- There's also a massive class imbalance. Unhappy customers who churned are heavily outnumbered by those who stayed. If I don't fix this, a predictive model will just lazily guess "No" every time and look artificially accurate. I'm deploying SMOTE (Synthetic Minority Over-sampling Technique) to upsample the minority class, forcing a mathematically symmetrical 50/50 balance.
- Visualizations using Matplotlib and Seaborn will prove the data's integrity before and after these transformations.

By the end of this pipeline, the output will be a perfectly sanitized, balanced dataset. It sets a concrete foundation for the final exam, where this data will feed directly into predictive models to accurately red-flag high-risk accounts before they pull the plug.